

Welcome



2025

DSC DONOR RE-ACTIVATION: A DATA-DRIVEN APPROACH

INFURNA Beatrice maria,
TRAN Minh Hoan,
NGUYEN Duc Manh



Business

Introduction.

DSC aims to re-activate inactive donors by sending physical letters encouraging them to donate again. The primary objective is to identify candidates likely to contribute **€30 or more**. Currently, sending letters to the entire database is costly and inefficient. The management requires proof that a predictive model performs significantly better than a random selection.

With a fixed campaign cost of **€0.80** per letter, we must demonstrate the model's value by comparing:

- **Baseline:** The performance of selecting donors randomly (no model).
- **Model:** The performance of targeting specific donors ranked by their likelihood to donate.

Business Question:

“How can DSC utilize donor data to outperform random selection and maximize campaign profit?”

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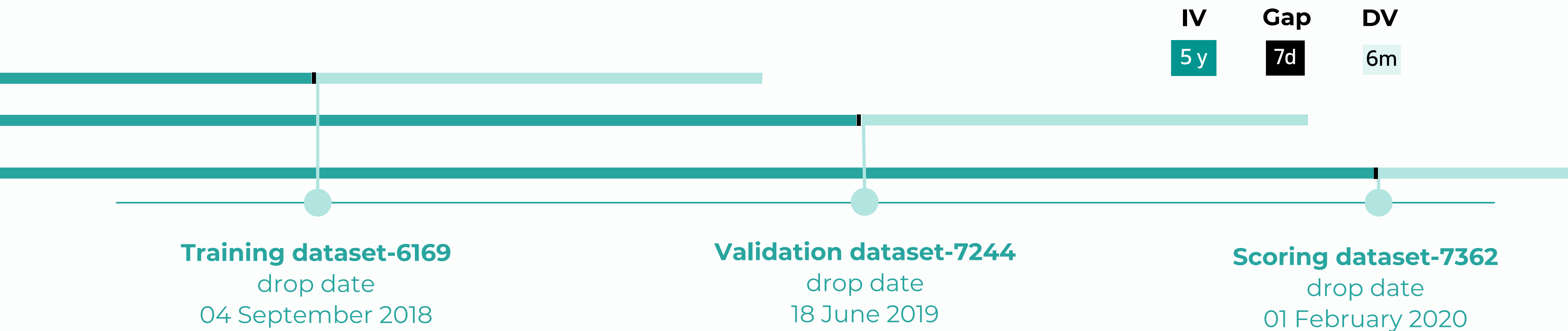
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Business Case
& Conclusion

Timeline and features



70+

Tested features

To deliver the best quality

Demographic & Static Features

Demographic, Donation
Summary, Campaign
participation

Behavioral Donation Patterns

Recency, Frequency,
Streaks, Trends, Volatility

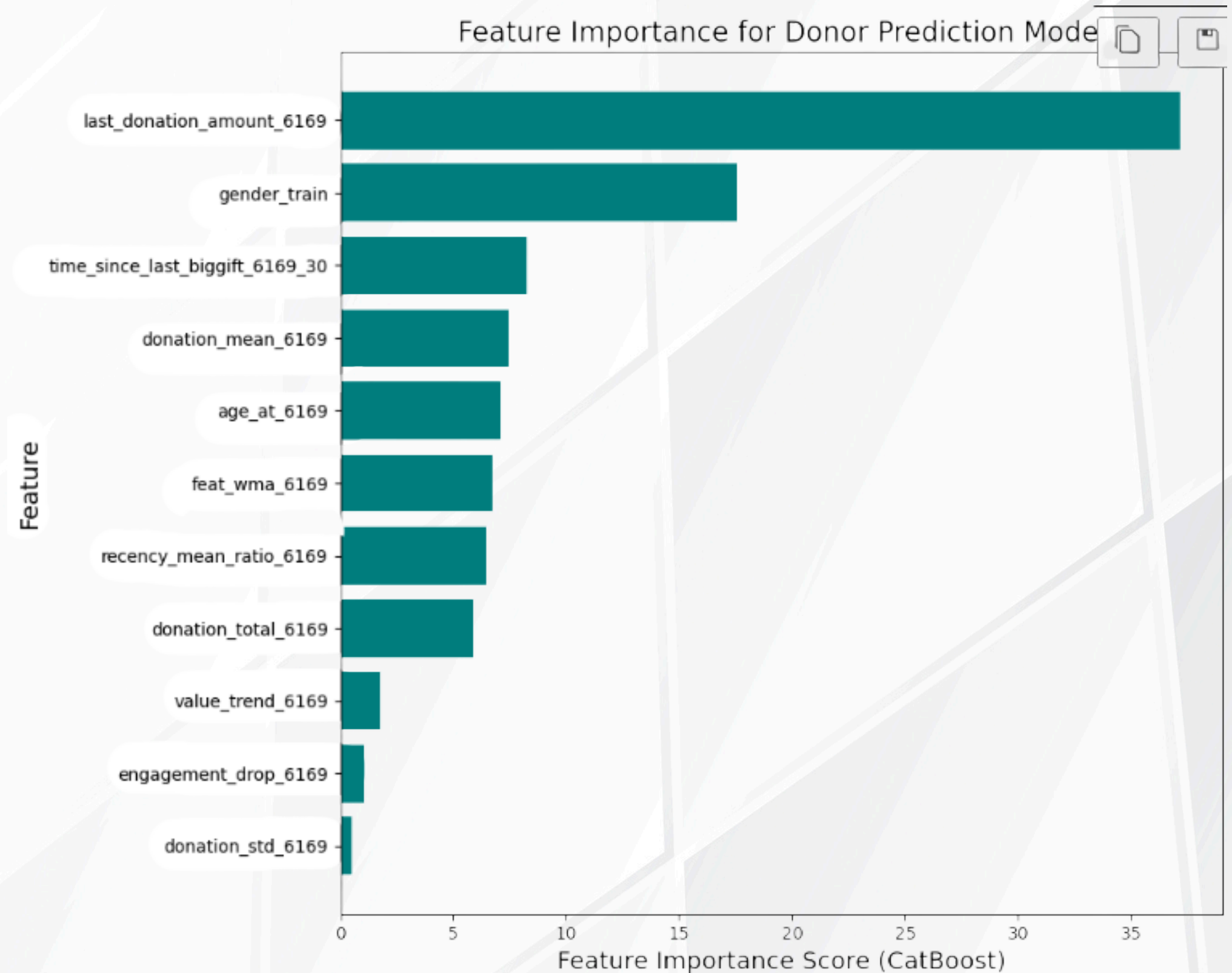
Long-memory / Reactivation Indicators

Ideal for a reactivation
campaign.

io

FEATURE SELECTION

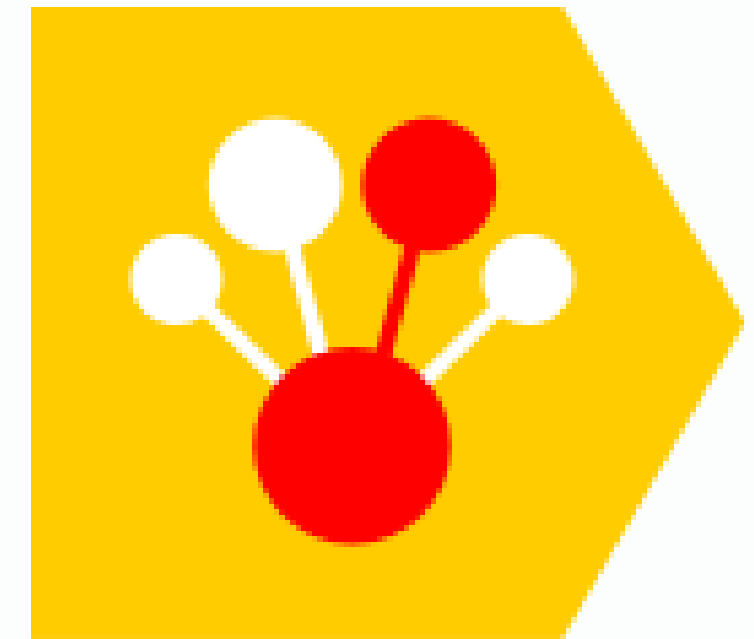
- **Removed highly correlated** features to avoid redundancy.
- **Retained features with stronger business logic** (donation behavior > metadata).
- Applied **Mutual Information Selection**, which performs well on **imbalanced datasets**.
- Used an **elbow-based threshold (0.0040)** to keep only features with meaningful predictive power.
- Final selection: **11 features** out of 70+, improving model stability and reducing overfitting.



Model Performance & Evaluation

--- AUC-ROC and Accuracy Comparison ---

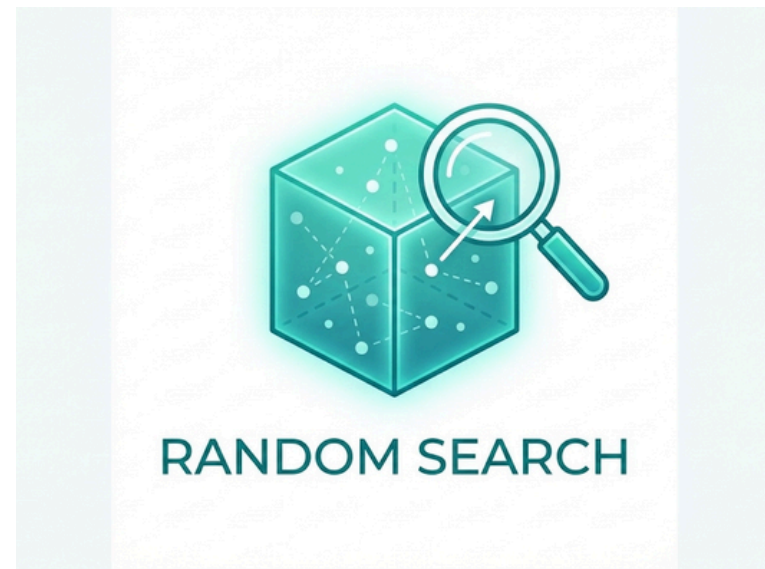
Model	AUC-ROC	Accuracy
CatBoost	0.6958	0.7968
Random Forest	0.6887	0.7901
LightGBM	0.6864	0.8015
XGBoost	0.6697	0.7620
Gradient Boosting	0.6509	0.7057



Cat boost

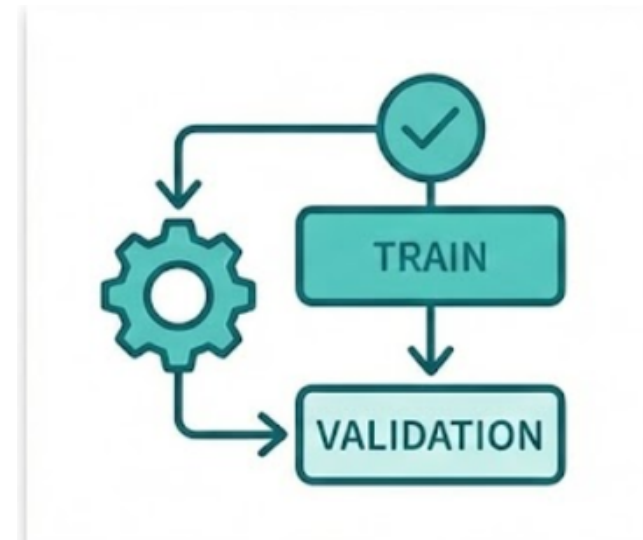
- Handles categorical features automatically
- Robust against overfitting
- High performance on tabular data
- Can handle imbalance data

Tuning the model



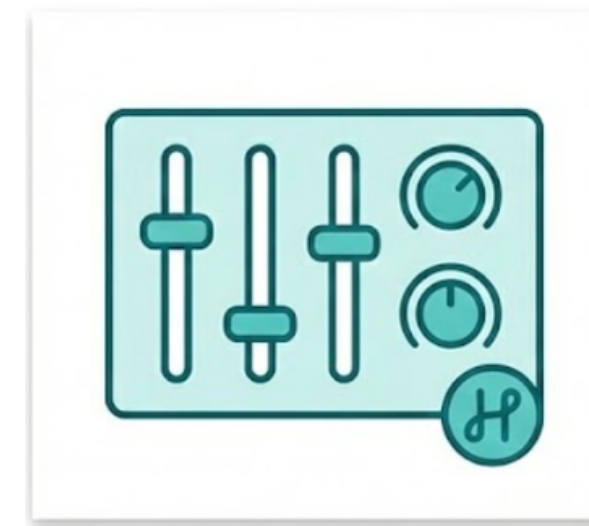
Random search

Covers a wider range of parameter combinations.



Validation Strategy

Train on Resampled Data vs. Validate



Hyperparameters

Tree Depth, Learning Rate, L2 Regularization



Threshold Moving

Maximize Accuracy on imbalanced data

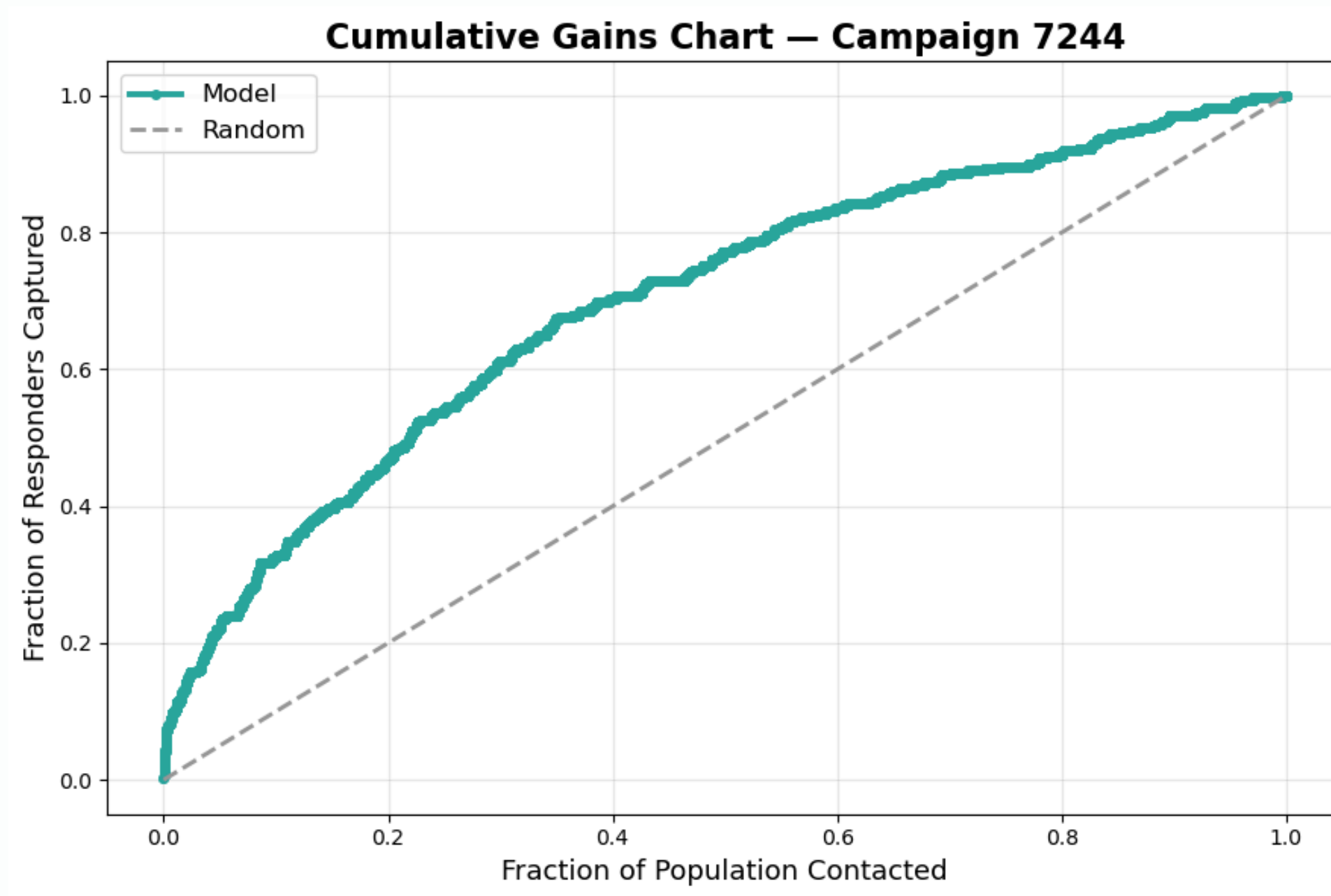


BEST VALIDATION AUC: 0.7017



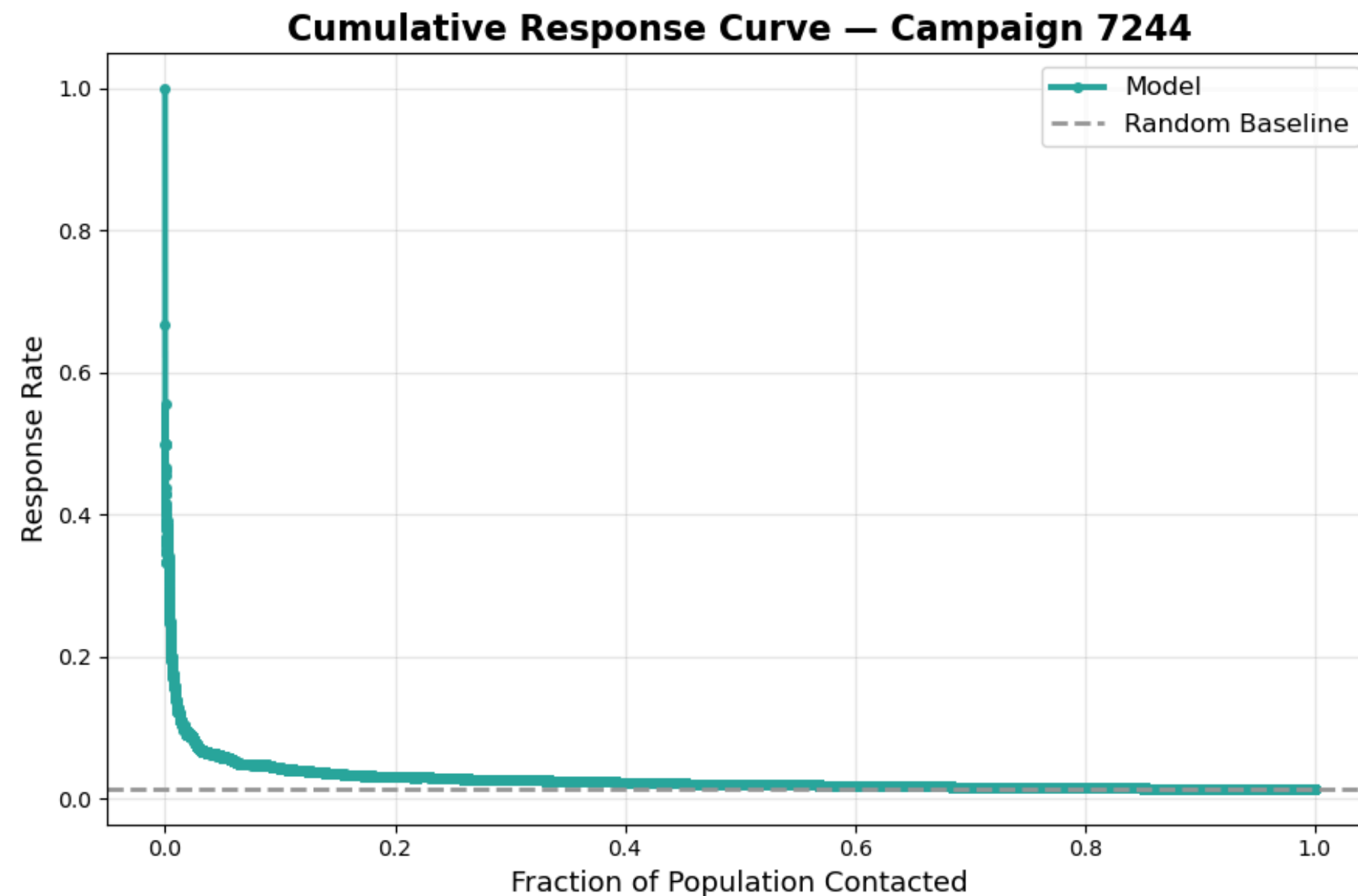
Best Accuracy: 0.9297 at Threshold: 0.75

Cumulative Gains Curve



- **Horizontal Axis (X-axis):** Percentage of total donor capture
- **Vertical Axis (Y-axis):** Percentage of population contacted
- Shows how much of the total target audience we capture.
- **Insight:** By targeting the top **30-40%** of scored customers, we can capture nearly **60-70%** of all donors.

Lift Curve

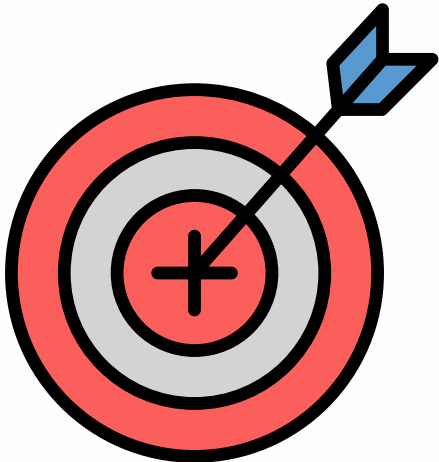


- **Horizontal Axis (X-axis):** Percentage of Selected Customers (0% to 100%).
- **Vertical Axis (Y-axis - Lift):** Effectiveness (Multiples of the Average).
- Shows how much better the model is compared to random targeting.
- **Insight:** In the top deciles (**top 10-20%**), the model is nearly **1.5 to 2 times** more effective than random selection.

Business Case for Campaign 7244

Objective: Evaluate our model-based targeting profitability vs random targeting in campaign 7244

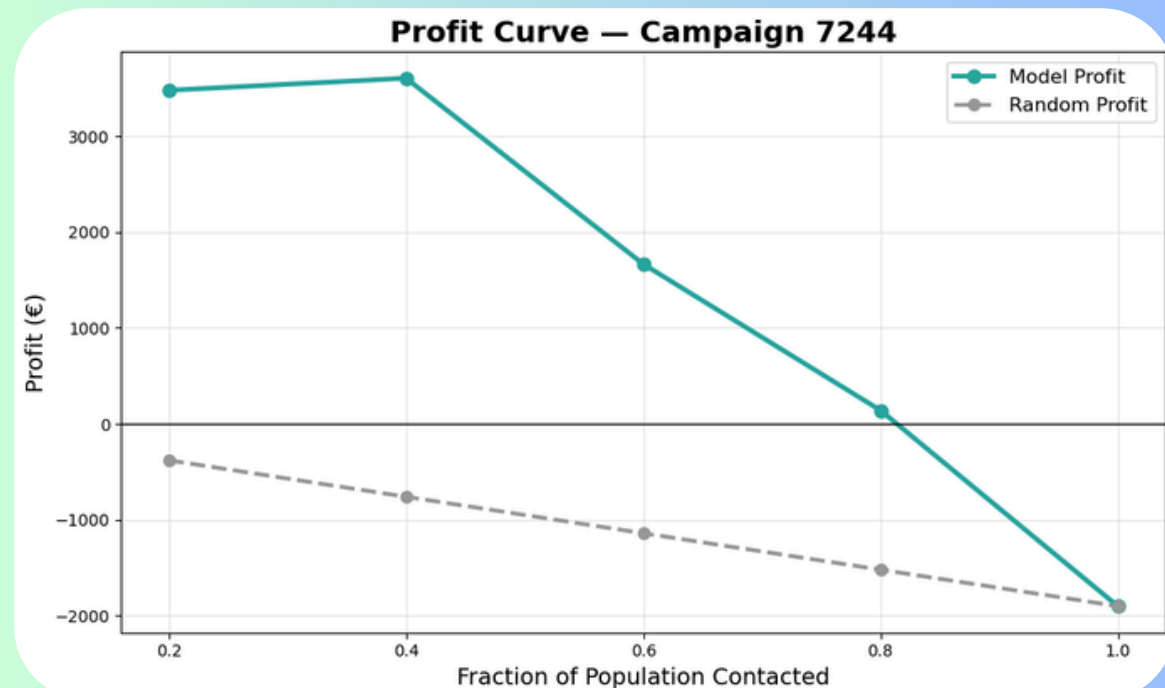
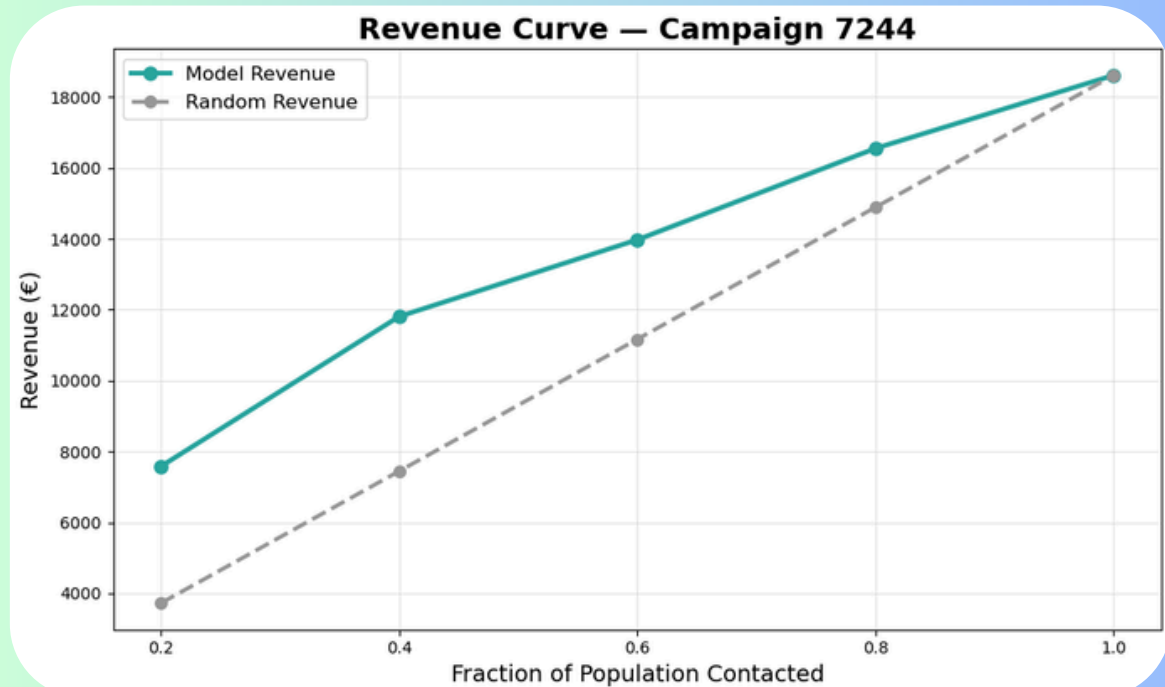
Campaign Setup	
Mailing cost per donor	0,80€
Average donation	51,56€
Base campaign response rate	1,41%
Total number of donors selected	25645



Goal

- Determine:
- Whether our model is profitable
 - What is the optimal contact rate
 - How much additional revenue and profit the model generates

Revenue & Profit Results (Model vs Random)



- **Model outperforms random** at every fraction.
- **Random selection** is always **unprofitable** (negative profit).
- Model becomes profitable immediately **at 20%**.
- Peak **at 40%**, with **revenue of 3600**, diminishing returns after.
- **Mailing 100%** of donors results in a **loss for both strategies**.

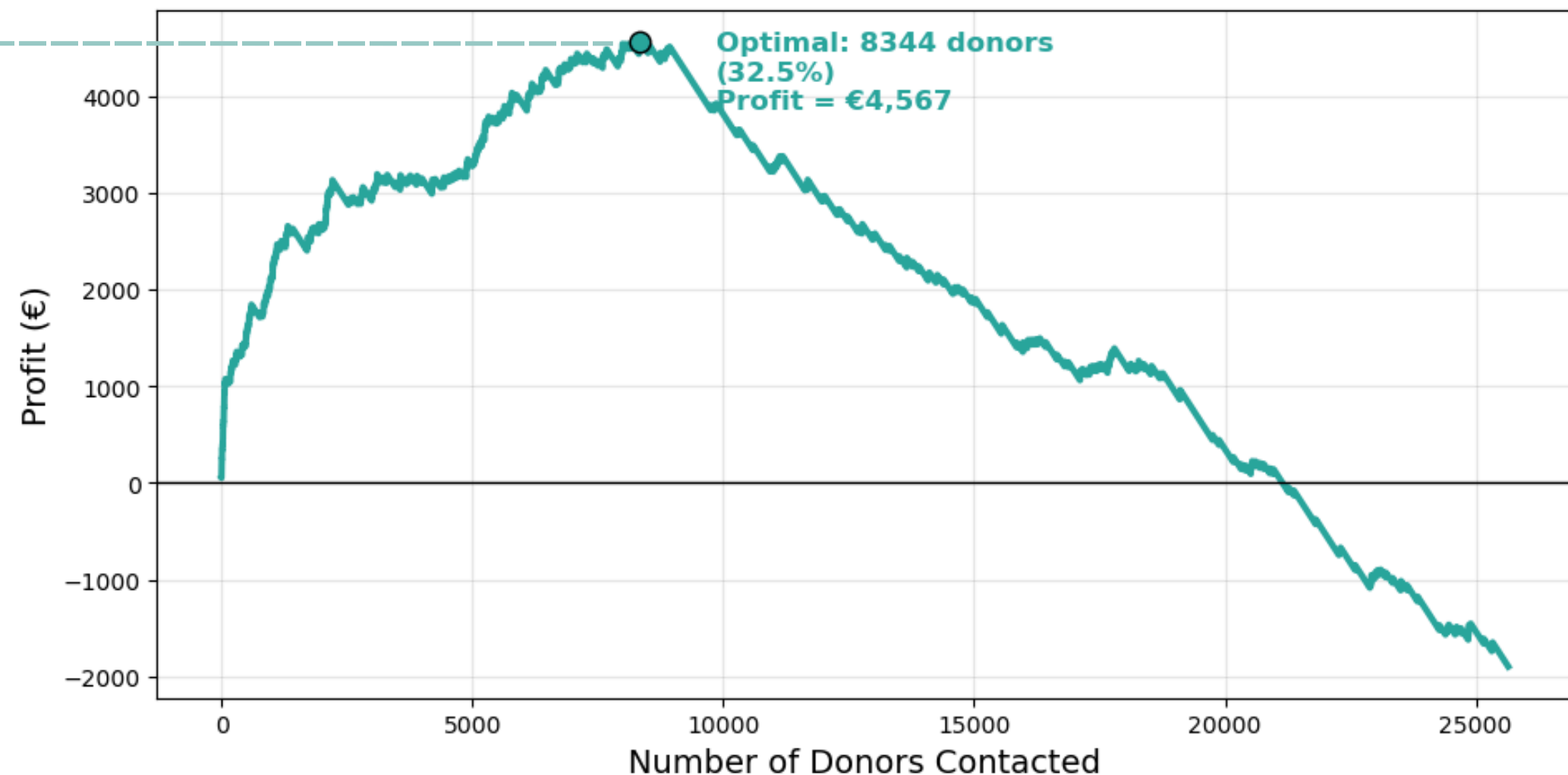
Fraction Contacted	Revenue-Model	Profit- Model	Revenue-Random	Profit- Random
20%	7.580,5	3.477,3	3.723,2	-380,0
40%	11.809,0	3.602,6	7.446,4	-760,0
60%	13.974,9	1.665,3	11.169,6	-1140,0
80%	16.553,3	140,5	14.892,8	-1520,0
100%	18.616,0	-1900,0	18.616,0	-1900,0

| Optimal Contact Strategy

~33%
Optimal contact
threshold

- The model identifies the **highest-value donors early**.
- Profit peaks at **32.5%** of donors contacted, **€4,567 profit**.
- Contacting more than **~8,300 donors** results in diminishing returns;
- After ~40–50%, additional contacts reduce profit
- Mailing the **full list** results in **a loss**

Full Profit Curve — Optimal Contact Fraction

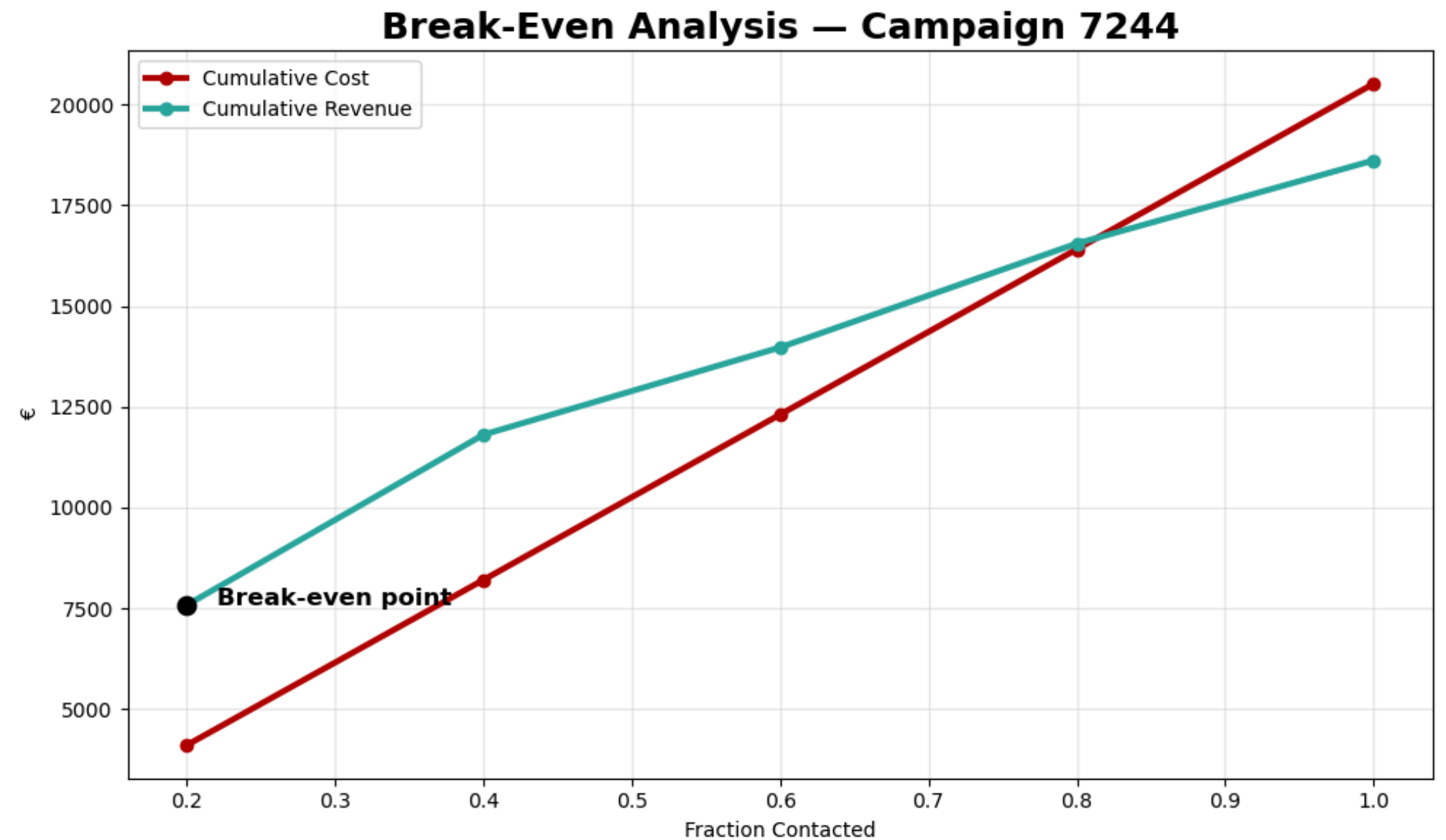


Business Impact – How Much Value the Model Creates

The model turns an unprofitable campaign into a profitable one

- **Random targeting** → –€1,900 loss
- **Model-based targeting** → €4,567 profit
- **Total improvement:** +€6,467
- **Cost savings:** € 13,800

ROI
+68% At **33%** Contact Level.
Every **€1** invested yields **€1.68** back.



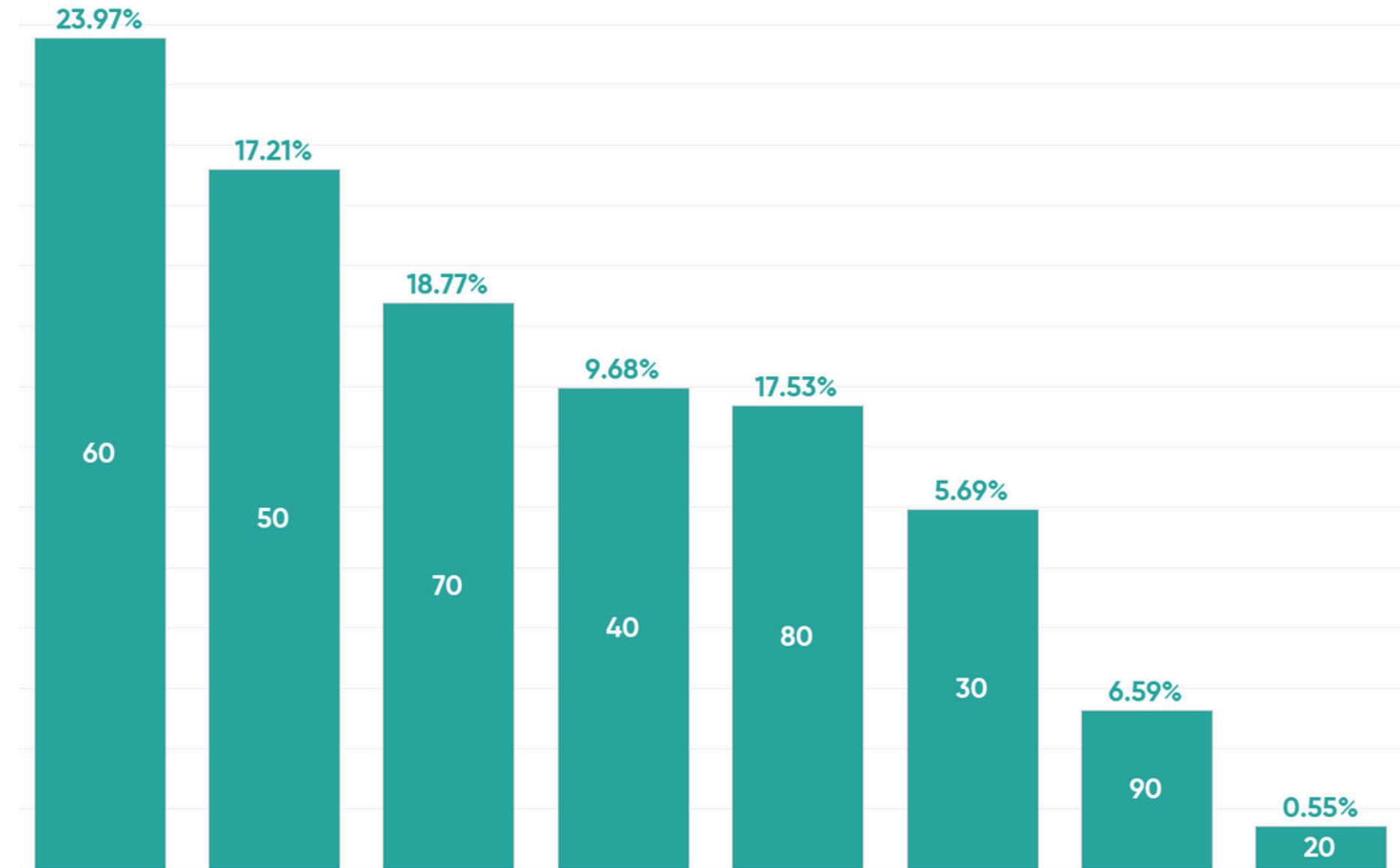
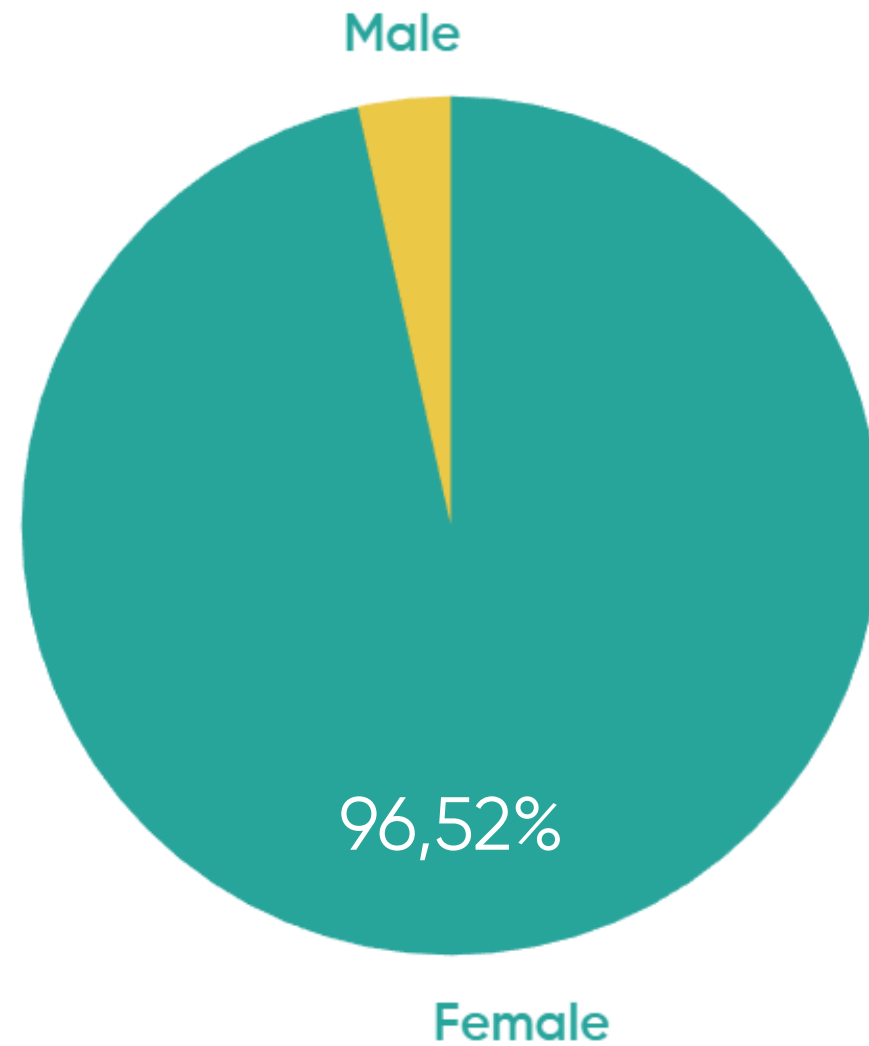
Predictive targeting increases fundraising profitability by more than €6,400 compared to business-as-usual.

Donor Profiling

"Who are your likely donors? What is their profile?"



| Demographics: The Senior Segment



Average Age: ~62.3 years

Core Range: 50–75 years

Gender: Overwhelmingly Female (96.5%)

| Giving Behaviour Indicators



Retention

82% of high-scoring donors show
no significant drop in
engagement history.



Stability

Value trend is flat (0), indicating
consistent donation levels rather
than wild fluctuations.



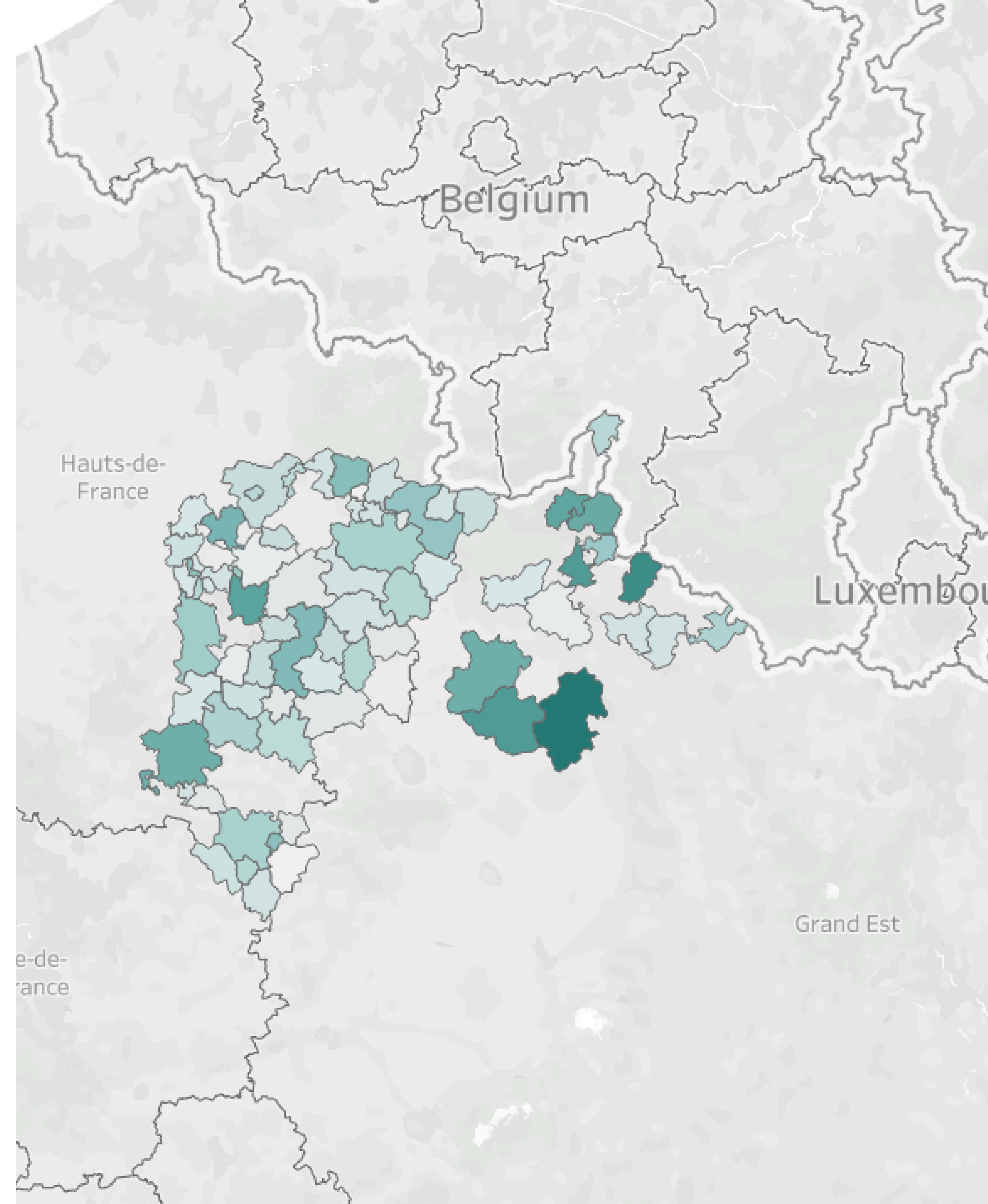
Activity

**Average Weighted Monetary
Activity (WMA) is ~17.3**, signaling
consistently active donors.

REGIONAL INSIGHT

There is a strong concentration of predicted donors in:

- **Southern Belgium**
- **Central and Eastern Belgium (Wallonia region)**
- Several communes near the **Luxembourg border**



| Model Segmentation

The predictive model identifies a distinct "Ideal Donor" profile with high precision:

- **Gender:** Female
- **Age:** Mature Adults (60–75 years)
- **Lifecycle Stage:** Transitioning to retirement or recently retired.
- **Behavior:** Stable engagement history
- **Prediction:** High Probability Score (85%)
- **Strategy:** Small, consistent gifts create the biggest foundation.

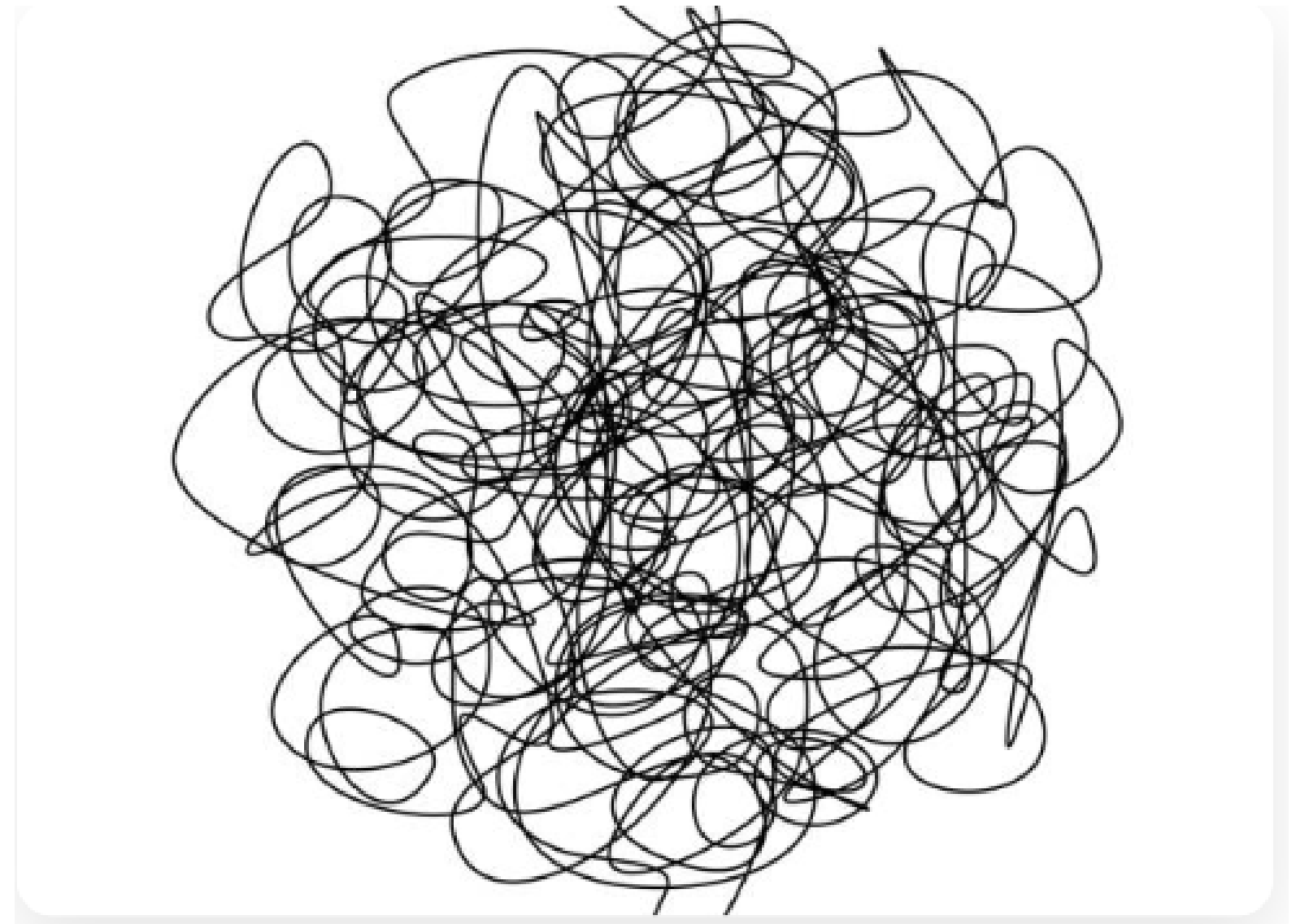


SUMMARY & NEXT STEPS

The Challenge: Random Selection

Without a targeted approach, the campaign suffers from significant waste:

- **Negative ROI:** The campaign currently loses money.
- **High Costs:** Marketing expenses exceed the donation intake.
- **Low Response:** Only ~29% of contacted individuals respond.



The Solution: Predictive Scoring

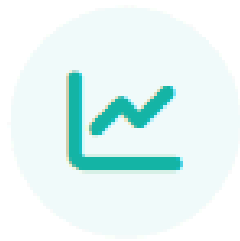
33%

Target Audience

The "Sweet Spot" Strategy:

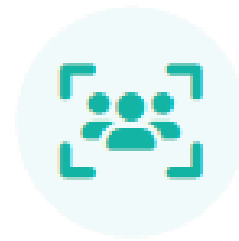
- By shifting from random selection to a predictive model, we dramatically change the outcome.
- Our analysis identifies that targeting only the Top 33% of scored donors turns the campaign from a loss into a profitable venture, filtering out non-responders before money is spent.

Financial Impact & Efficiency



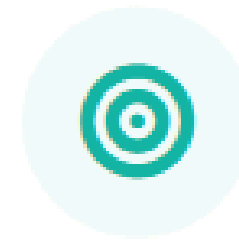
Maximized ROI

The optimized strategy yields a return of approximately **€4,567**, reversing previous losses.



2x Efficiency

The model finds almost **twice as many responders** for the same budget compared to random targeting.



Precision

Resources are focused strictly on high-probability donors, eliminating wasted spend.

Next Step: Execution

The model is validated and ready for deployment.

Action Item: Apply the predictive model to the final scoring dataset for Campaign 7362.





THANK YOU



Do you have any questions?
